

```
28.
29. if len(data) == 0:
30.     print(f"No data found for {ticker}. Please check the ticker symbol.")
31.     return None
32.
33. print(f"Retrieved {len(data)} data points.")
34.
35. # Handle missing values
36. missing_values = data.isna().sum().sum()
37. if missing_values > 0:
38.     print(f"Found {missing_values} missing values. Filling with forward fill method...")
39.     data = data.ffill().bfill() # Forward fill, then backward fill any remaining
40.
41. # Add useful derived columns
42. # Daily returns
43. data['Returns'] = data['Adj Close'].pct_change()
44.
45. # Volatility (20-day rolling standard deviation of returns)
46. data['Volatility'] = data['Returns'].rolling(window=20).std()
47.
48. # Trading volume relative to 20-day average
49. data['Relative Volume'] = data['Volume'] / data['Volume'].rolling(window=20).mean()
50.
51. # Remove rows with NaN values resulting from calculations
52. data = data.dropna()
53.
54. # Check for outliers in returns
55. z_scores = abs((data['Returns'] - data['Returns'].mean()) / data['Returns'].std())
56. outliers = z_scores > 3
57. outlier_count = outliers.sum()
58.
59. if outlier_count > 0:
60.     print(f"Detected {outlier_count} outlier days in returns.")
61.     # Note: We're not removing them, just flagging for awareness
62.     data['Is Outlier'] = outliers
63.
64. print("Data preparation complete.")
65. return data
66.
```